

PPS-2(LAB-0)

QUE-1: Primality Test

```
⇒ #include <cmath>
#include <cstdio>
#include <vector>
#include <iostream>
#include <algorithm>
using namespace std;
```

```
int isPrime(int n){
    if(n<=1) {
        return 0;
    }
    for(int i=2;i*i<=n;i++){
        if(n%i==0){
            return 0;
        }
    }
    return 1;
}
```

```
int main() {
    /* Enter your code here. Read input from STDIN. Print output to
    STDOUT */
    int number;
    cin >> number;
    for(int i=0; i<number; i++) {
        int temp;
        cin >> temp;
        if(isPrime(temp)) {
```

```

        cout << temp << " ";
    }
}
return 0;
}

```

QUE-2: Decimal Integer to Binary

```

⇒ #include <cmath>
#include <cstdio>
#include <vector>
#include <iostream>
#include <algorithm>
using namespace std;

```

```

int main() {
    /* Enter your code here. Read input from STDIN. Print output to
    STDOUT */

```

```

    int num;
    cin >> num;
    int maxi_num = 1;

    while(maxi_num <= num) {
        maxi_num *= 2;
    }
    maxi_num /= 2;
    while(maxi_num != 0) {
        cout << num/maxi_num;
        num %= maxi_num;
        maxi_num /= 2;
    }

```

```
    return 0;
}
```

QUE-3: Print 15 leap years from a given year

```
⇒ #include <cmath>
#include <cstdio>
#include <vector>
#include <iostream>
#include <algorithm>
using namespace std;
```

```
int main() {
    int year;
    cin >> year;

    if(year % 100 != 0) {
        if(year % 4 == 0) {
            int n=15;
            do{
                cout << year << " ";
                year += 4;
                n--;
            } while(n != 0);
        }
        else{
            if(year%2 ==0 ) {
                year = year + 2;
            }
            if(year%2 != 0) {
                year -= 1;
            }
            int n=15;
            while(year%4==0 && n!=0) {
                int temp = year + 4;
```

```

        cout << temp << " ";
        year += 4;
        n--;
    }
}
}
else {
    if(year % 400 == 0) {
        int n=15;
        do{
            cout << year << " ";
            year += 4;
            n--;
        } while(n != 0);
    }
    else {
        int n=15;
        while(year%4==0 && n!=0) {
            int temp = year + 4;
            cout << temp << " ";
            year += 4;
            n--;
        }
    }
}

return 0;
}

```

QUE-4: Reversing a number

```

⇒ #include <cmath>
#include <cstdio>
#include <vector>
#include <iostream>

```

```
#include <algorithm>
using namespace std;
```

```
int main() {
    int n;
    cin >> n;
    int rem, var=0;
    while(n!=0) {
        rem = n %10;
        var = var * 10 + rem;
        n/=10;
    }
    cout << var;
    return 0;
}
```